Review: Angle relationships

1
a No b Yes c No d No

2
a No b Yes c No d No

3
a Yes b No c No d No

4
a Yes b No c No d No

5
a No b No c No d Yes

6 180°

7
a No b Yes c No

8
a Yes b No

9
a 51 b 42

10
a 97 b 86

11
a $x = 60$ b $y = 72$
c $z = 10$

12

a $x=70$

b $x=25$

c $x=20$

d $x=35$

e $x=75$

f $x=40$

g $x=40$

h $x=14$

i $x=27$

j $x=\dfrac{61}{3}$

k $x=18$

13 Example answers: $\angle AXD, \angle AXS, \angle PXD, \angle PXS, \angle BXC, \angle BXR, \angle QXC, \angle QXR$

14

a Example answer: $\angle DXE$ and $\angle EXF$, $\angle AXB$ and $\angle BXC$, $\angle AXB$ and $\angle AXF$, $\angle BXC$ and $\angle CXD$, $\angle CXD$ and $\angle DXE$

b $\angle BXC$ and $\angle EXF$

c $\angle AXB$ and $\angle BXC$

15 Example answer: $\angle CXD, \angle CXS, \angle RXD, \angle RXS$

16

a $x=22$

b 68°

17

a $x=28$

b 54°

18

a $x=28$

b 33°

19

a $x=45$

b $y=135$

20

a No

b No

c Yes

d Yes

21 $m\angle TAP=32^{\circ}$

Additional questions

22

a These angles are not complementary angles because $68^\circ + 12^\circ = 80^\circ$ and complementary angles must add up to 90°.

b These are vertical angles because they are opposite angles formed by intersecting lines and have equal measures.

c These angles form a linear pair since they are adjacent and $109^\circ + 71^\circ = 180^\circ$.

d These angles do not form a linear pair since $157^\circ + 43^\circ = 200^\circ$ and a linear pair must add up to be 180°.

23

a $\angle QON$

b $\angle AEI$

24

a $a = 27$

b $m = 51$

25

a Answers can include $\angle TWU$ and $\angle TWY$, $\angle VWU$ and $\angle VWY$, $\angle YWZ$ and $\angle ZWU$, or $\angle YWX$ and $\angle XWU$

b There are no vertical angles in the diagram

c $\angle YWV$ and $\angle VWT$

d $\angle TWU$ and $\angle YWX$

26

a

i $10x + 1 = 12x - 5$

ii Vertical Angles Theorem: vertical angles are congruent so their measures are equal

iii $x = 3$

b

i $4x + 7 + 2x + 5 = 180$

ii Linear Pair Postulate: angles that form a linear pair are supplementary so the sum of their measures is equal to 180°

iii $x = 28$

27

a $x = 30$

b $x = 23$

28 45°

29 10°

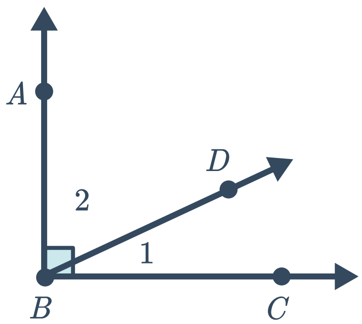
30

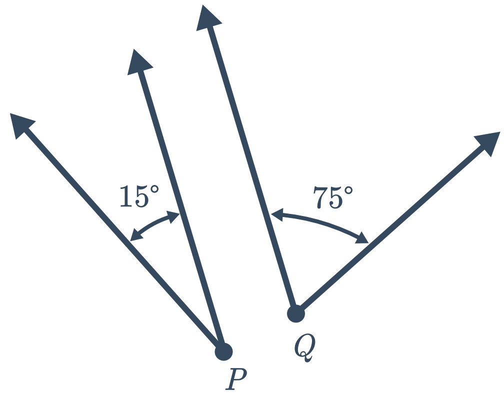
a 30°

b 150°

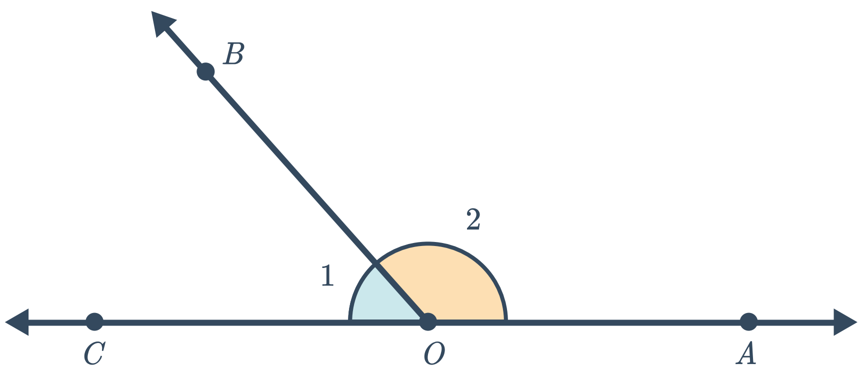
31

a Sometimes

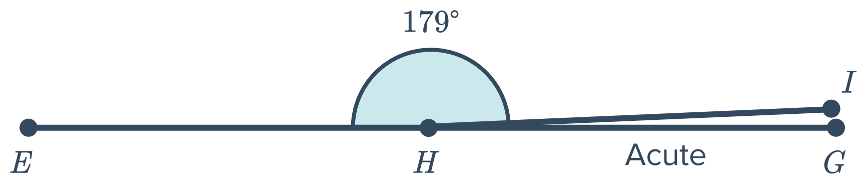


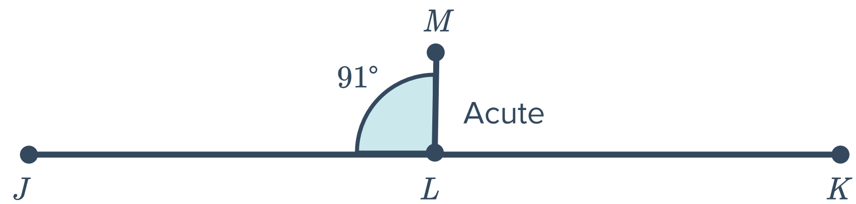


b Always. A linear pair forms a straight angle and a straight angle always measures 180°

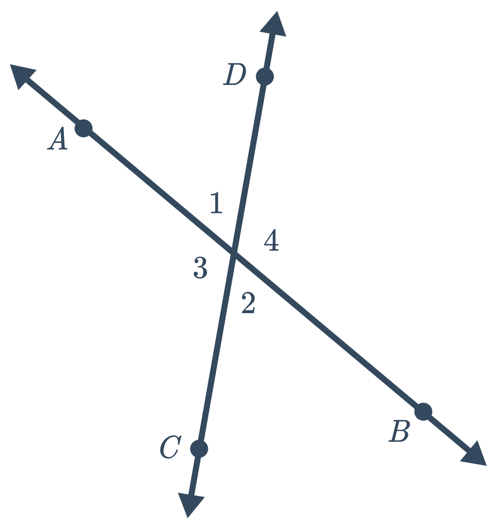


c Always. Consider the extreme measures [redacted]tuse angles: An obtuse angle measuring on the smaller end at 91 degrees, the linear pair angle can be found by subtracting $180-91=1^{\circ}$ which produces an acute angle and an obtuse angle measuring on the larger end at 179 degrees has a linear pair with the measure $180-179=1^{\circ}$

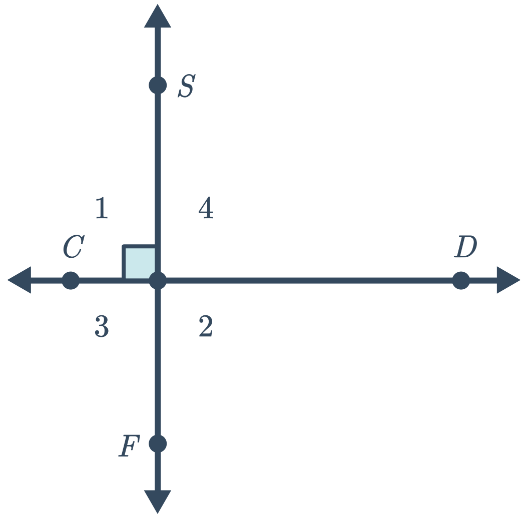




d Never. Vertical angles are the opposite angles from intersecting lines and can never be adjacent. To form a linear pair angles must be adjacent.

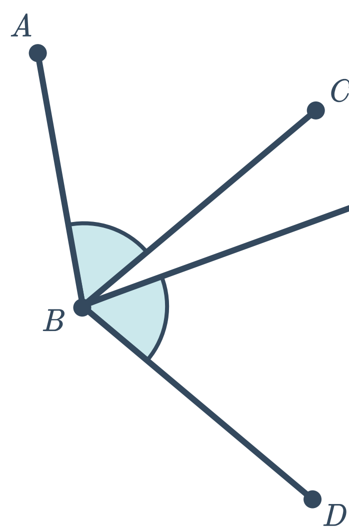


e Sometimes. If the vertical angles are 90° they will also be supplementary.



32

a This is not enough information to make the conclusion. The angles share a common vertex and are congruent, but vertical angles are only formed from intersecting lines. Here is a counterexample:



b This is enough information to justify the conclusion. Since $\angle MNP$ and $\angle PNR$ form a linear pair we know they are supplementary and we can write $m\angle MNP + m\angle PNR = 180$. Let $x = m\angle MNP$. We also know that $\overline{NP}$ bisects $\angle PNR$ which makes $\angle MNP \cong \angle PNR$ by definition of angle bisector. So we can conclude that $x = m\angle PNR$ too. Substitute x into our supplementary equation and we get $x + x = 180^\circ$ which means $x = 90^\circ$.